

# Tour Length Comparison - 150-City Maps

Only tour length is shown. Source 100-200 values are sorted within each panel by BlindBlendACO tour length: 1-101.

Algorithm colors

BaselineACO

BlindBlendACO

ElitistACO

MaxMinACO

Tour length averages

With Cross Check

BaselineACO: 29,612

BlindBlendACO: 28,370

ElitistACO: 29,493

MaxMinACO: 28,541

Without Cross Check

BaselineACO: 30,383

BlindBlendACO: 29,012

ElitistACO: 30,094

MaxMinACO: 29,415

With Cross Check

This line chart displays the tour length for 101 sorted maps/runs across four algorithms: BaselineACO (blue), BlindBlendACO (purple), ElitistACO (red), and MaxMinACO (green). The y-axis, labeled 'Tour length', ranges from 25,928 to 33,675. The x-axis, labeled 'Sorted map / run index', ranges from 1 to 101. Each algorithm's performance is represented by a line with circular markers. Horizontal dashed lines indicate the average tour length for each algorithm. BaselineACO shows the highest variability and generally the highest tour lengths, while BlindBlendACO shows the lowest variability and generally the lowest tour lengths.

| Algorithm     | Average Tour Length |
|---------------|---------------------|
| BaselineACO   | 29,612              |
| BlindBlendACO | 28,370              |
| ElitistACO    | 29,493              |
| MaxMinACO     | 28,541              |

Without Cross Check

This line chart displays the tour length for 101 sorted maps/runs across four algorithms: BaselineACO (blue), BlindBlendACO (purple), ElitistACO (red), and MaxMinACO (green). The y-axis, labeled 'Tour length', ranges from 25,928 to 33,675. The x-axis, labeled 'Sorted map / run index', ranges from 1 to 101. Each algorithm's performance is represented by a line with circular markers. Horizontal dashed lines indicate the average tour length for each algorithm. The overall tour lengths are higher than in the 'With Cross Check' scenario, with BaselineACO showing the highest variability and generally the highest tour lengths, and BlindBlendACO showing the lowest variability and generally the lowest tour lengths.

| Algorithm     | Average Tour Length |
|---------------|---------------------|
| BaselineACO   | 30,383              |
| BlindBlendACO | 29,012              |
| ElitistACO    | 30,094              |
| MaxMinACO     | 29,415              |

Sorted map / run index